

MÉDICO YACOUBA DIOMANDE

Energy Data Scientist | Infrastructure Intelligence Specialist

Predictive Analytics · AI Agents & Automation · Donor-Funded Electrification

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What I Bring: Early Warnings, Trusted Numbers, Follow-Up That Runs Itself

Time. Delays spotted early: a delay-risk model (**93% recall**), progress alerts across 55 localities, and an agent (live demo) that runs the weekly follow-up of stalled worksites on its own.

Serenity. Numbers you can trust: field experience separates real signals from data-entry noise (two real anomalies caught), and SHAP shows why each forecast says what it says.

Credibility. Reporting that holds up with donors and project units: World Bank compliance, an audited tracking database with role-based access, structured technical reports.

Why bet on me: 10+ years on donor-funded electrification (Head of Engineering since 2024), RNCP Level 7 Data Scientist, and four public projects (3 live demos, 1 platform in production).

Next step: a 30-minute call to map where this applies to your programme.

CORE COMPETENCIES

AI & ML	Supervised learning (classification, regression), clustering & PCA, time-series forecasting, explainability (SHAP), imbalanced data (SMOTE, threshold tuning)
AI Agents & Automation	LangGraph (stateful agents, SqliteSaver persistence), LangChain, n8n workflows, FastAPI · Also covered: CrewAI, AutoGen/AG2, BeeAI (IBM/Coursera certification)
Platform Engineering	AppSheet, Google Sheets, Apps Script: multi-tenant SaaS, role-based security, alerts
MLOps, Dev & Viz	MLflow, GitHub Actions (CI/CD), FastAPI, Flask, Streamlit, pytest, Docker (basic), Python, SQL, Git · Power BI, Matplotlib, Seaborn, Plotly
DL & Big Data	NLP (BERT, TF-IDF, Word2Vec), Computer Vision (CNN), PySpark, AWS (S3, EC2, EMR)
Engineering & PM	HV/LV network design, feasibility studies, APD, tender files (DAO), site supervision · World Bank compliance, scheduling, progress KPIs, risk management

PROFESSIONAL EXPERIENCE

Head of Engineering Department | Afrique Ingénierie et Experts Consultants (AIEC) | Apr 2024 – Present

- ▶ Lead multi-disciplinary teams on large-scale rural and urban electrification projects financed by the World Bank.
- ▶ Embed data analysis and automation in project delivery to anticipate delays and sharpen forecasting.
- ▶ Ensure technical and budget compliance across concurrent projects, with data-backed reporting to funders.

For you: one accountable lead keeping engineers, field teams and funders on the same reliable numbers.

Project Engineer — Energy Infrastructure | AIEC | 2016 – Mar 2024

- ▶ Designed and validated HV/LV network feasibility studies and detailed execution designs (APD) for CI-Énergies.
- ▶ Managed cross-functional teams on 6+ World Bank and AFD-funded electrification programmes.
- ▶ Turned field data into technical reports and investment files; coordinated procurement and quality control.

Selected programmes

- ▶ **PTDAE** (World Bank, CI, 2019–2022): full project lifecycle, from technical validation to stakeholder reporting.
- ▶ **12 regional capitals** (AFD, 2018–2019): tender-ready package (field data, feasibility study, APD, tender file).
- ▶ **85 HV/LV substations, Abidjan** (CI-Énergies, 2019–2020): LV reinforcement designed across all 85 (APD).

For you: studies, designs and tender files to World Bank and AFD standards, from field data to procurement.

Earlier roles: Independent Study Expert, SIDEES (UEMOA solar PV feasibility studies), 2016 · Technical Supervisor, Afrique Energie Services, 2014–2015.

KEY PROJECTS — DATA SCIENCE & AI (PORTFOLIO, 2026)

Public portfolio: github.com/Medcos/Energy-Data-Scientist · Pseudonymised or synthetic data; no client identifiers.

Predictive Rollout & Supply Forecast — *Rural electrification · 55 localities · pseudonymised real project data*

- ▶ **What it brings you:** advance warning on material shortfalls, slipping progress and delay risk, so you act before the delay, not after.
- ▶ Three-model suite: material-forecast error cut **~2–3×** vs. no model (7 vs. 20 units); progress at +4 weeks within **4.3 points** (7.3 naive); delay-risk **recall of 93%** (15 of 55 localities at risk).
- ▶ Baselines first (Gradient Boosting kept in 1 of 4 cases, on evidence); SHAP on every model; two real data-entry anomalies caught through field knowledge.

Stack: Python · pandas · scikit-learn · SHAP · Streamlit | [Live demo](#) · [Code](#)

ElecTrack Pro — Multi-Tenant Field Operations Platform — *in production on 55 localities*

- ▶ **What it brings you:** one governed place for field progress, materials and automated alerts, where each role sees only its own zone.
- ▶ Weighted progress, materials reconciliation, GPS-aware logging, scheduled snapshots; data-integrity rules; a one-workbook-per-client (v2) architecture built for multi-client resale.

Stack: AppSheet · Google Sheets · Google Apps Script · Role-Based Security | [Code & demo](#)

Jobsite Relaunch Agent — n8n + LangGraph — *Agentic automation · synthetic data derived from a real project*

- ▶ **What it brings you:** stalled worksites followed up every week without manual spreadsheet checks.
- ▶ **9-node stateful LangGraph agent:** typo-tolerant locality matching with ambiguity detection, execution-pace calculation, trend-aware status classification, week-over-week memory persisted in SQLite.
- ▶ FastAPI service orchestrated by a weekly n8n workflow; Streamlit simulator; **41 automated tests**.

Stack: Python · LangGraph · n8n · FastAPI · SQLite · Streamlit · pytest | [Live demo](#) · [Code](#)

Predictive Maintenance for Industrial Machines — *Two-level ML · public benchmark dataset (10,000 obs.)*

- ▶ **What it brings you:** likely failures, and their probable cause, flagged before they stop production.
- ▶ Failure detection: Gradient Boosting, **AUC-ROC 0.98, F1 0.88** (precision 0.93, recall 0.82); root cause: F1 of 0.94–1.00 on three failure modes, with per-mode thresholds.
- ▶ Physics-informed features (power = torque × speed, thermal differential), validated by SHAP.

Stack: Python · scikit-learn · SHAP · Streamlit | [Live demo](#) · [Code](#)

KEY PROJECT — ENGINEERING

P2AE — Power Access Augmentation Project — *World Bank · Bénin · Oct 2024 – Present*

- ▶ **Field delivery:** supervise HV/LV rural electrification works across 55 localities in four departments: technical files and execution studies validated, progress tracked against contractual milestones, inspection teams coordinated.
- ▶ **Live visibility:** designed and run the project's digital tracking layer (ElecTrack Pro): weighted advancement per locality, weekly snapshots, automated alerts on stalled sites.
- ▶ **Data you can defend:** audited the tracking database (referential integrity, access control) and delivered a prioritised remediation roadmap; the most critical security findings have since been fixed.

EDUCATION, CERTIFICATIONS & LANGUAGES

Building AI Agents and Agentic Workflows — Specialization | IBM / Coursera | Aug 2026

Agent frameworks: LangGraph, CrewAI, AutoGen, BeeAI. Applied in the Jobsite Relaunch Agent project. [Verify credential](#)

Master's-Level Data Scientist — RNCP Level 7 | OpenClassrooms / CentraleSupélec, Paris | 2022 – 2025

Remote programme; equivalent to a French Master's degree (NSF 326), accredited by France Compétences.

Electromechanical Engineer Degree | ESEP — Abidjan, Côte d'Ivoire | 2008 – 2016

Certification projects: credit scoring with MLOps deployment (LightGBM, SHAP, MLflow, CI/CD) · sentiment analysis at scale (PySpark, AWS EMR) · product classification (NLP + vision) · customer segmentation (RFM, K-Means).

Further training: IBMI Berlin — project, risk & change management, leadership (2018–2021) · Smart Grid Specialist · Energy Transition & Efficiency (French universities, 2017–2019).

Languages: French (native) · English (professional, fluent) · Arabic (intermediate)